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CHAPTER ATA 77

ENGINE INDICATING

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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Engine Indicating – General

Introduction

The purpose of the indicating system is to provide data on vital engine parameters for use by the flight crew and maintenance personnel to monitor engine health and performance. These parameters are: High Pressure Rotor Speed (NH), Low Pressure Rotor Speed (NL), Propeller Speed (NP), Torque (Q), Air Intake Temperature (T1.8), and Inter-Turbine Temperature (ITT).

General Description

[Refer Figure - Engine Indicating System – Block Diagram](#) & [Engine Indicating Details](#)

The engine parameters collected by the sub-systems are shown on the Engine Display. The parameters are also sent to the EMU for use by the maintenance personnel when they access the maintenance screen on the ARCDU or the CDS on the laptop.

The function of the indicating system is performed by the:

- Power indicating sub-system
- Temperature indicating sub-system
- Analyzers

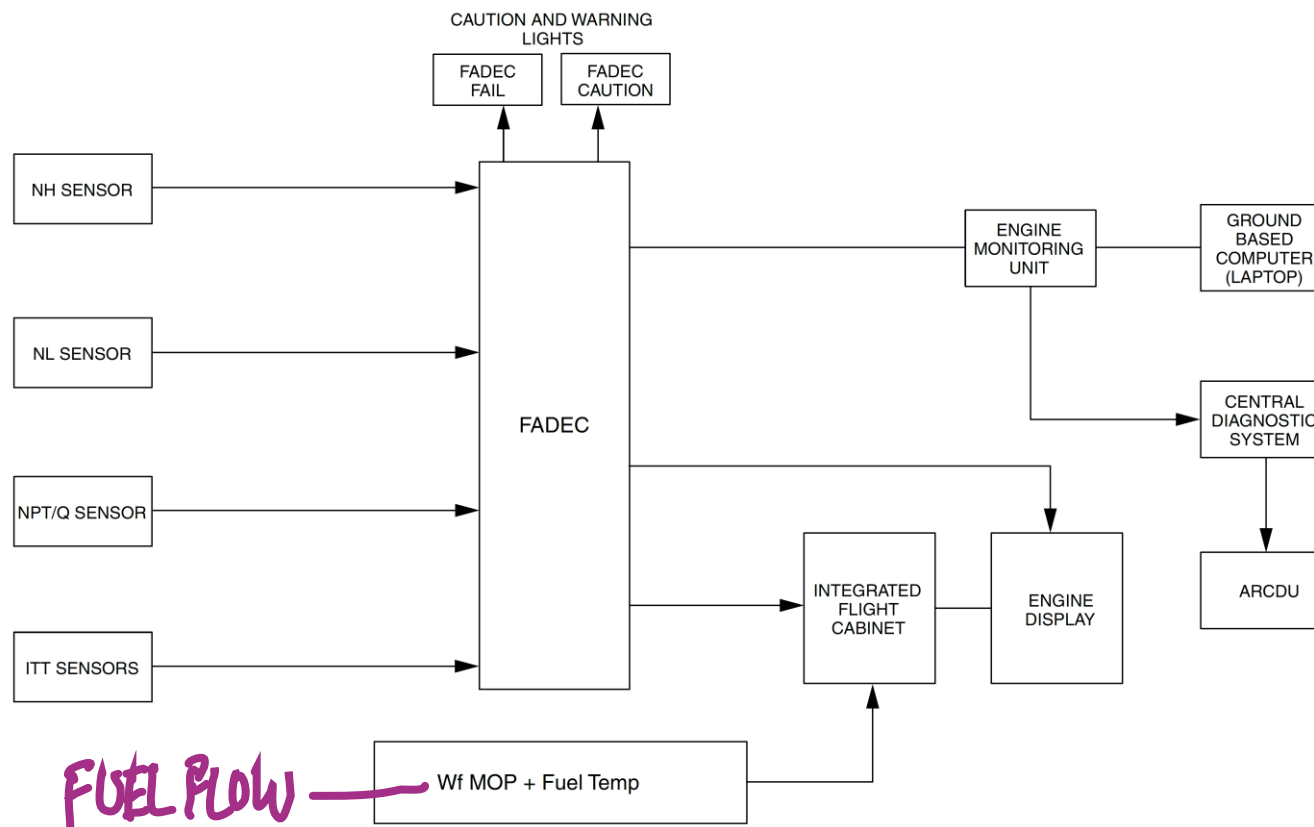
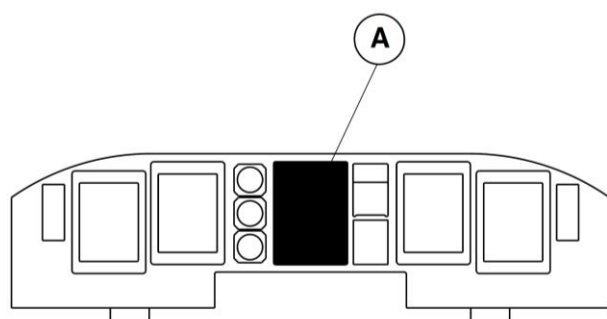


Figure 77-1 Engine Indicating System – Block Diagram



MAIN INSTRUMENT PANEL

LEGEND

1. Fuel Quantity and Fuel Temperature Indications.
2. Balance Message Indicating Fuel Imbalance.
3. Autofeather Armed Indication.
4. Torque Bug.
5. Uptrim In Operation.
6. Left/Right Torque Indication.
7. Engine Mode Indication.
8. Torque Bug Digital Readout.
9. High Pressure Gas Generator Speed.
10. Overspeed Governor Test In Progress.
11. Fuel Flow In Pounds Per Hour.
12. Indicated Turbine Temperature.
13. Low Pressure Gas Generator Speed.
14. Oil Pressure and Temperature Indications.

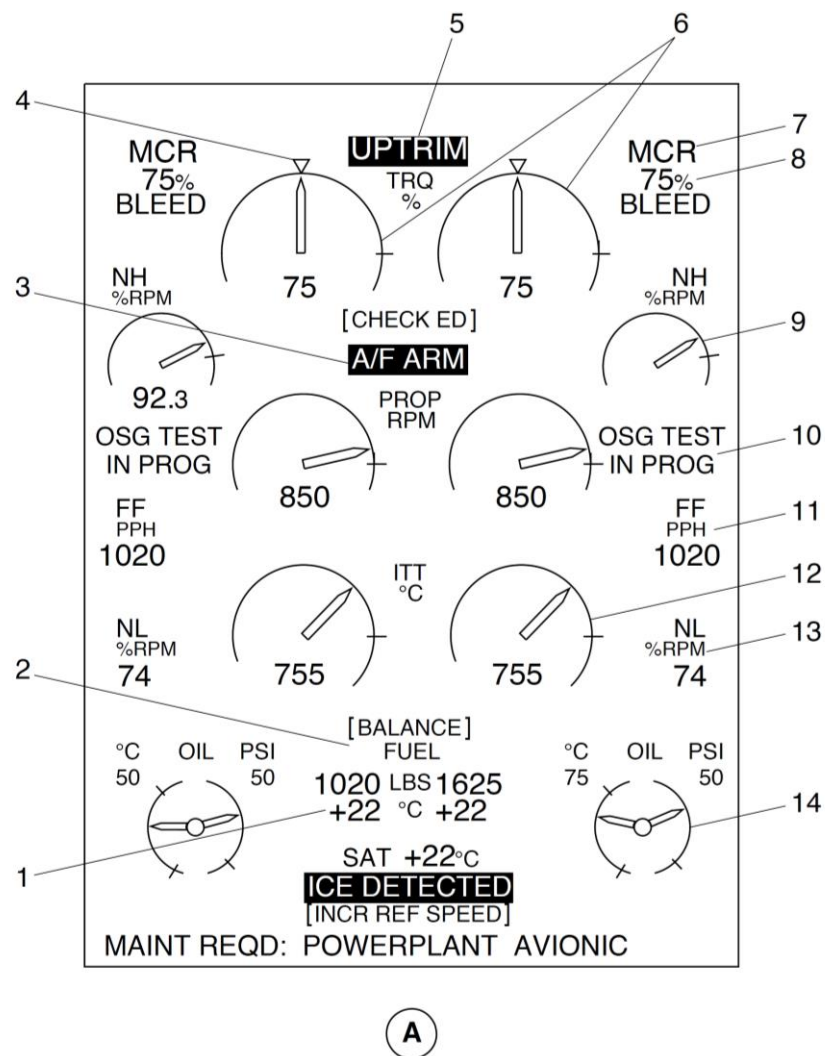


Figure - Engine Indicating Details

Engine Power Indicating

Introduction

The Power Indicating System provides data on the engine high pressure rotor speed (NH), torque (Q), and low pressure rotor speed (NL).

General Description

[Refer Figure - Engine Power Indicating Schematic](#)

The NH, Q, and NL sensors provide signals which are passed to the Full Authority Digital Electronic Control (FADEC) by the controls, electrical wiring harness. Each FADEC display channel (channel not in control) sends the signals on an ARINC 429 bus direct to the Engine Display.

The power indicating system has the components that follow:

- High Pressure Rotor Speed Sensors (NH)
- Torque Sensors (Q)
- Low Pressure Rotor Speed Sensor (NL)

Power Indicating System

The engine parameters collected by the sub-systems are shown on the Engine Display. The parameters are also sent to the Engine Monitoring Unit (EMU) for use by the maintenance personnel when they access the maintenance screen on the Audio Radio Control Display Unit (ARCDU) or the Central Diagnostic System (CDS) on the laptop.

Detailed Description

The speeds of the major rotating assemblies are monitored by magnetic pulse pick-up probes installed at various locations on the engine. High pressure rotor speed (NH), Low pressure rotor speed (NL), and propeller speed (NP) are sensed by these probes. Power turbine rotor speed (NPT) is sensed by the torque sensor and derived to propeller speed (NP). Electromagnetic pulses are generated when associated gear teeth or lugs pass through the magnetic field created at the probe or sensor tip. The pulse frequency is transmitted to the Full Authority Digital Electronic Control (FADEC) and Propeller Electronic Control (PEC) and then to the Central Diagnostic System (CDS).

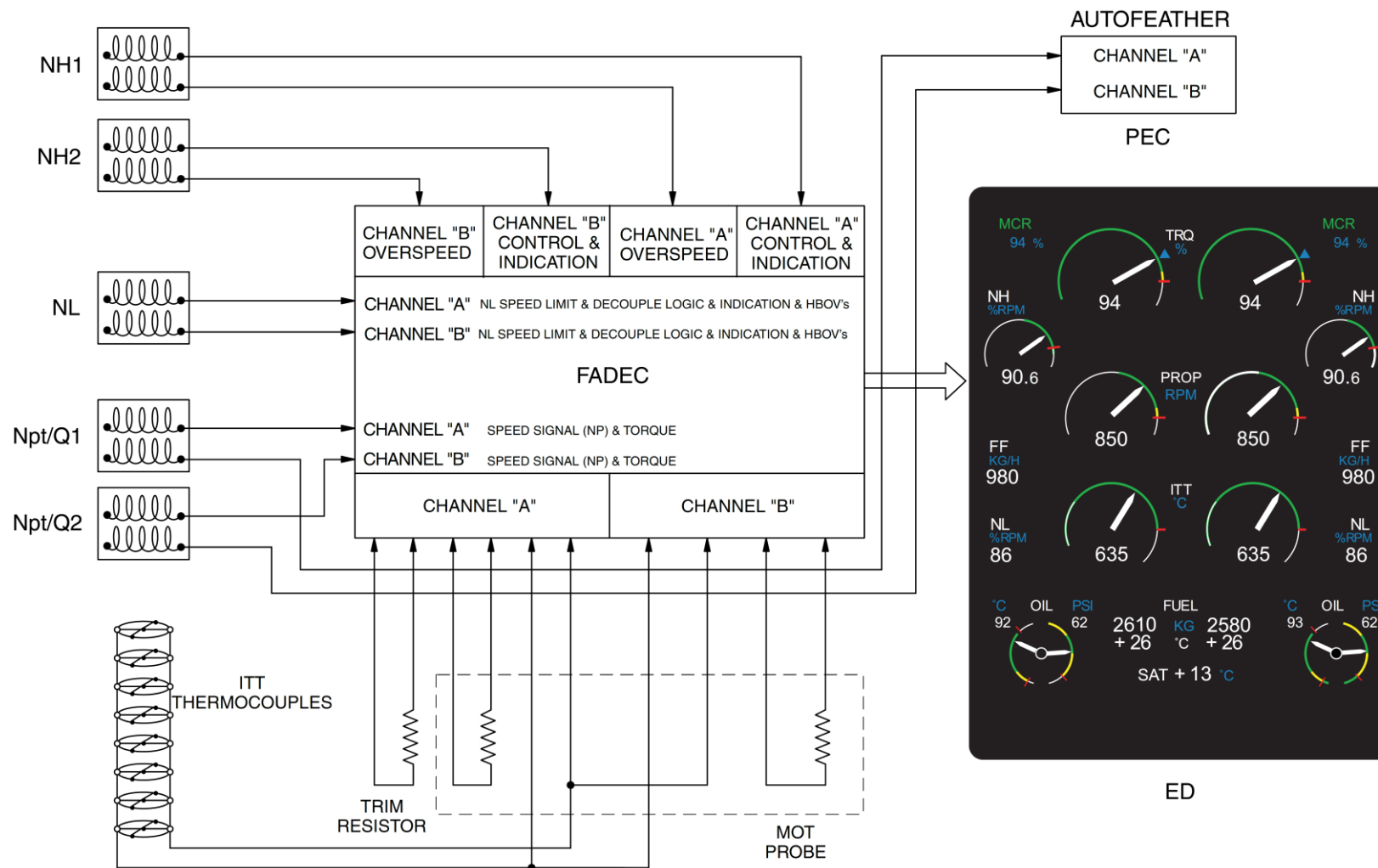


Figure - Engine Power Indicating Schematic

Engine Power Indicating – Component Description

High Pressure Rotor Speed (NH) sensors

Refer Figure - NH Sensor

There are two NH sensors located on top of the Accessory Gearbox (AGB). Each NH sensor is a sealed unit consisting of two coils. The sensors pick up high pressure rotor speed signals from the fuel pump and PMA gearshaft teeth. One coil is used to provide a NH speed signal to the FADEC for use in the control logic, the other is used to provide a NH speed signal for use in the overspeed logic. There are two sensors on each engine, therefore control and overspeed functions are independent for channel A and B.

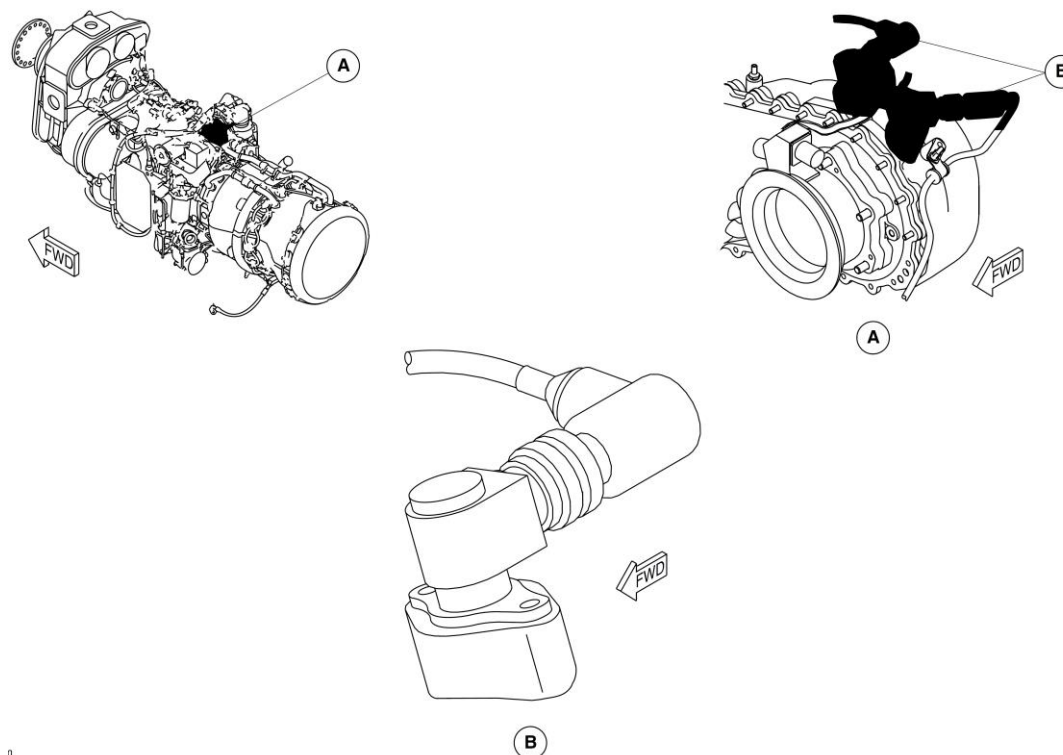


Figure - NH Sensor

Torque Sensor (Q)

Refer Figure - Torque Sensors

There are two torque sensors located on the air inlet case. The two torque sensors take their signal from torque shafts in the reduction gearbox. Correction of torque shaft stiffness caused by temperature change is provided to FADEC and PEC by the Main Oil Temperature (MOT) probe on top of the front inlet case. Each sensor consists of two coils which are used to sense Torque (Q) and Power Turbine Speed (NPT). Each sensor supplies a Torque and NPT speed signal to the FADEC and Engine Display, and Propeller Electronic Controller (PEC) using one coil for each. This means that there are independent Torque and NPT speed signals for channel A and B of both the FADEC and PEC. The Torque and NPT speed signals are used in the FADEC control logic. The FADEC provides processed Torque and NPT speed for flight compartment indication. In the PEC they are used for Autofeather logic processing.

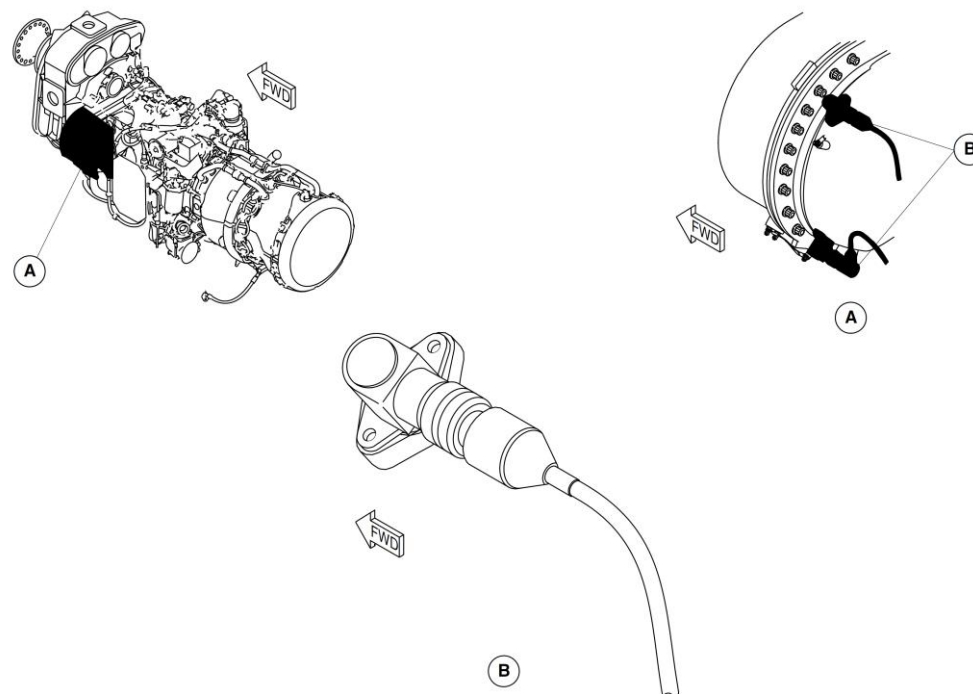


Figure - Torque Sensors

Low Pressure Rotor Speed Sensor (NL)

Refer Figure - NL Sensor

There is one Low Pressure Rotor Speed Sensor located on top of the front inlet case next to the MOT probe. The NL sensor is a sealed unit consisting of two coils. The sensor picks up low pressure rotor speed from a toothed sleeve against the no. 2.5 bearing. Each coil supplies a NL speed signal to channel A and B of the FADEC respectively. NL speed is used for controlling the Interstage and Intercompressor bleed valves, flight compartment indication and in the FADEC LP Rotor Overspeed Detection.

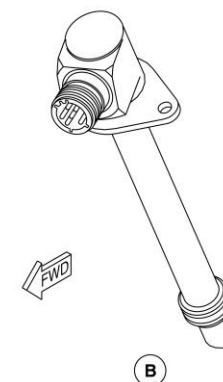
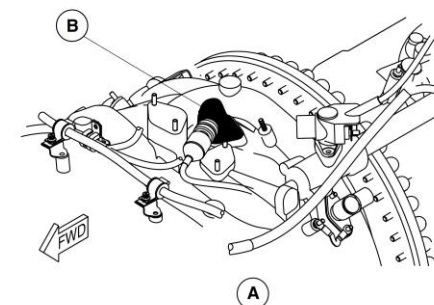
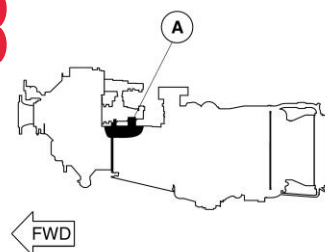


Figure - NL Sensor

Training Information Points

OBEY ALL THE ELECTRICAL/ELECTRONIC, AND ELECTROSTATIC DISCHARGE
SAFETY PRECAUTIONS

Engine Parameters Indication on Engine Display

[Refer Figure - Engine Display Indications](#)

The following indications are displayed in both digital and analog format on the ED as percentages:

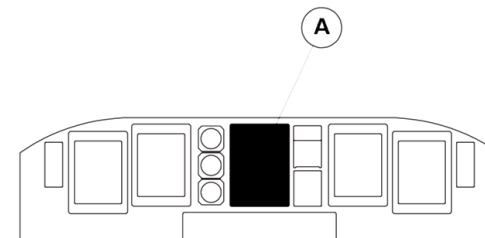
- High pressure compressor speed (Nh)
- Torque (Q)
- Propeller speed (Np)
- Inter Turbine Temperature (ITT)

Low pressure compressor speed (NL) is displayed in digital format only.

FADEC sends rating power information to the ED as indicated:

- By a torque bug
- On the torque analog display
- By digital display in the top left and right corners of the ED.

Selected Rating is indicated by a three letter acronym above the rated torque digital display.



MAIN INSTRUMENT PANEL

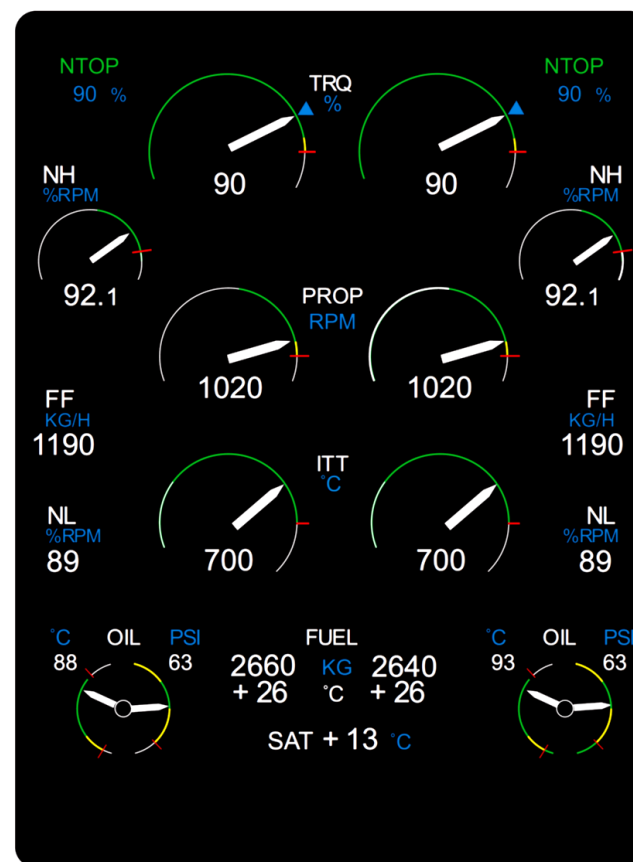


Figure - Engine Display Indications

Engine Temperature Indicating

Introduction

The Temperature Indicating System is used to monitor engine performance.

General Description

The temperature indicating system supplies data on internal engine temperatures at certain engine stations. **These temperatures are : Air Intake Temperature (T1.8), and Inter-Turbine Temperature (ITT).**

The function of the Temperature Indicating System is performed by the:

- Total-inlet temperature (T1.8) sensor

- Inter-turbine temperature probes

Detailed Description

[Refer Figure - ITT Indicating Schematic](#)

The T1.8 sensor and T6 (ITT) probes provide signals which are passed to the Full Authority Digital Electronic Control (FADEC) by the controls electrical wiring harness and the T6 wiring harness. Each FADEC display channel (channel not in control) sends the signals on an ARINC 429 bus direct to the Engine Display.

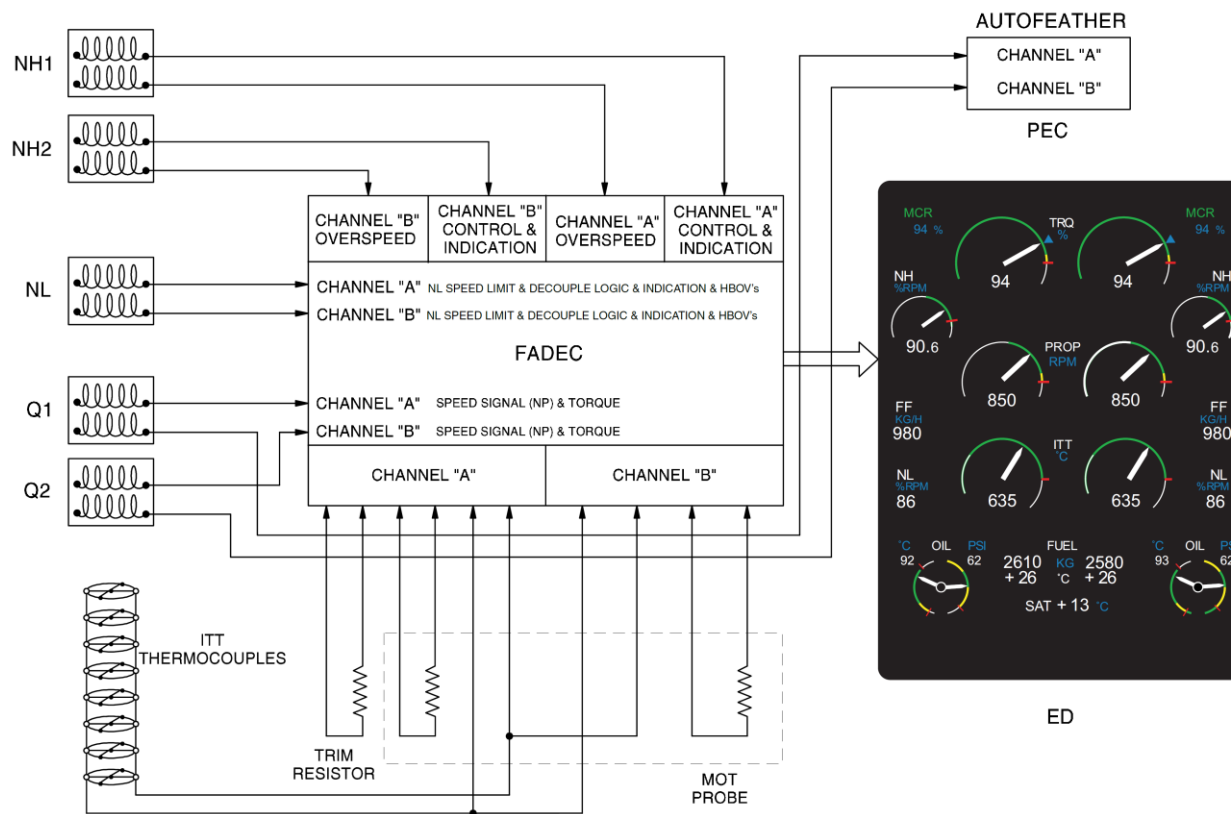


Figure - ITT Indicating Schematic

Engine Temperature Indicating – Component Description

T1.8 Temperature Sensor

Refer Figure - T 1.8 Temperature Sensor

A dual platinum resistance temperature device located in the intake just upstream of the first stage low pressure compressor. The sensor is used to measure air intake temperature. That temperature is used to calculate engine ratings. Calculated engine ratings are used to automatically set maximum rated power and display maximum rated torque in the flight deck.

Intake temperature measured by the T1.8 sensor is used by the local FADEC and transmitted to the opposite FADEC. The two values are averaged in each FADEC to ensure that Rated Torque displayed in the flight compartment, on the Engine Display (ED), is the same for both engines. T1.8 is the primary source of intake temperature to the FADEC. The maximum allowable difference between local and opposite T1.8 is ± 2 °C. The secondary source of intake air temperature is provided by the Air Data Unit (ADU).

The FADEC also monitors the ADU Static Air Temperature (SAT) from both channels. The SAT from the ADU could be used as primary source depending on the rating selected.

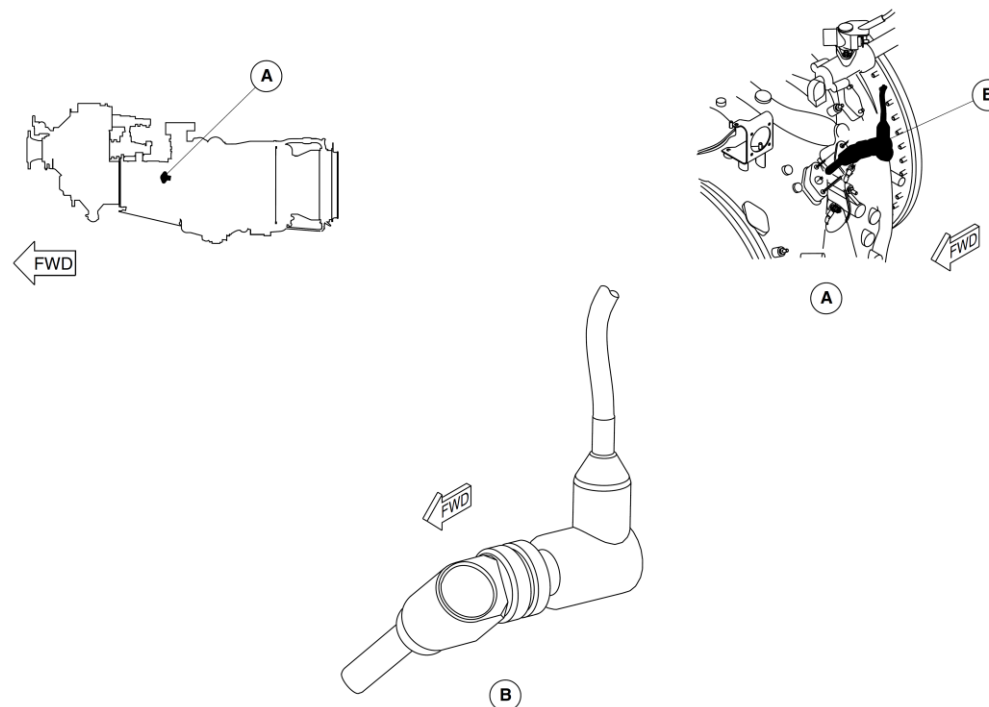


Figure - T 1.8 Temperature Sensor

Indicated Turbine Temperature (ITT) Probe

Refer Figure - ITT Thermocouples

Eight probes consisting of three temperature sensing elements provide 24 measurement points for comprehensive gas temperature measuring.

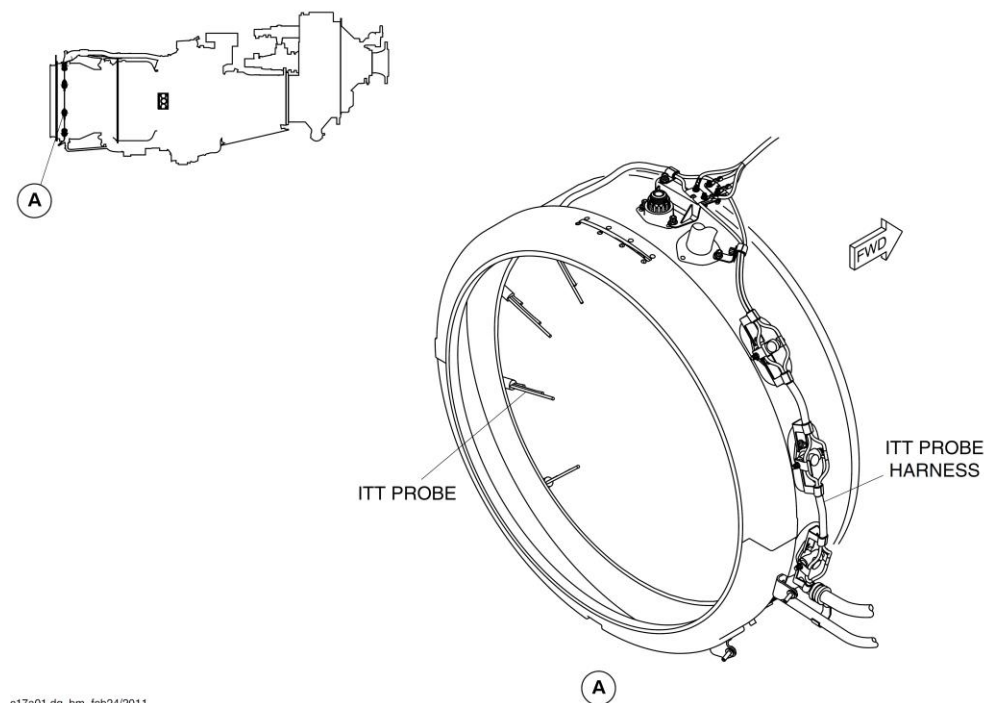
The sensing elements are located to provide sufficient coverage in order to calculate an accurate ITT. The probes are equi-spaced around the circumference at station 6 of the engine which is downstream of the 2nd Power Turbine.

ITT measurement is done by 24 sensing elements which are Chromel–Alumel thermocouples connected by the ITT wiring harness in parallel.

The total signal is connected at a junction located at the rear of the engine.

The junction is connected to a wiring harness which goes to the Main Oil Temperature/Cold Junction sensor.

Before being processed by the FADEC, the raw ITT is trimmed by the lug mounted resistor. The trimmed ITT and Main Oil Temperature (MOT) are processed by the FADEC and output to the flight compartment over ARINC 429. ITT is also sent to the Engine Monitoring Unit (EMU).



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Figure - ITT Thermocouples

ITT Trim Resistor

[Refer Figure - MOT Sensor and ITT Trim Resistor Detail](#)

The ITT Trim Resistor is located on two stud terminals on the Main Oil Temperature/Cold Junction Sensor.

ITT is produced from a system of 24 Chromel–Alumel thermocouples (eight sensors of three thermocouples each) arranged in parallel.

The average signal (by virtue of parallel arrangement) is routed through the ITT Trim Resistor and the cold junction reference before being received and processed by the FADEC. The ITT Trim Resistor is used to provide compensation to the raw ITT signal.

The value of the resistor is set during engine pass off testing at the manufacturer or approved overhaul facility.

Training Information Points

Replace failed Lug Mounted Resistors before replacement of the FADEC. The new FADEC will immediately ignore a failed resistor and use the trim value stored in its memory. This value will not be the correct value for the engine. Engine performance will be compromised.

OBEY ALL THE ELECTRICAL/ELECTRONIC, AND ELECTROSTATIC DISCHARGE SAFETY PRECAUTIONS WHEN HANDLING THE FADEC.

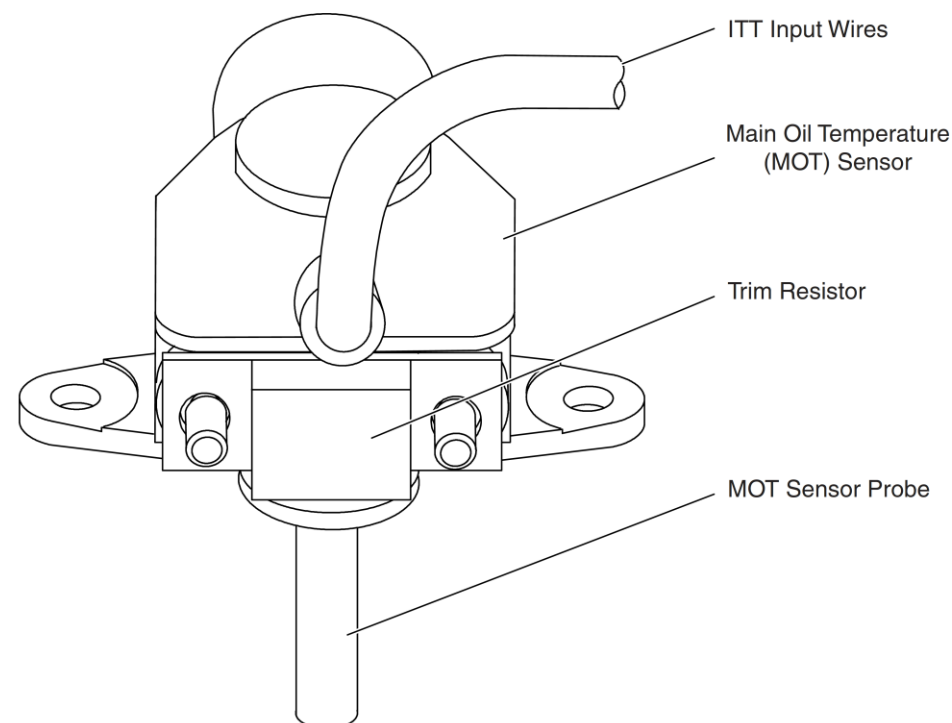


Figure - MOT Sensor and ITT Trim Resistor Detail

Analyzers

Introduction

This system is used to display most of the engine parameters, plus discrete indications, in the Flight Compartment.

General Description

[Refer Figure - EMS Block Diagram](#)

The analyzers record, display and download the information that follows:

- FADEC and PEC fault and condition codes.
- Engine Condition Trend Monitoring (ECTM) data.
- Power assurance requirements and data.
- Powerplant limit exceedance data.
- Snapshot and transient powerplant and aircraft data to help troubleshooting.
- Powerplant and aircraft flights, cycles and hours.

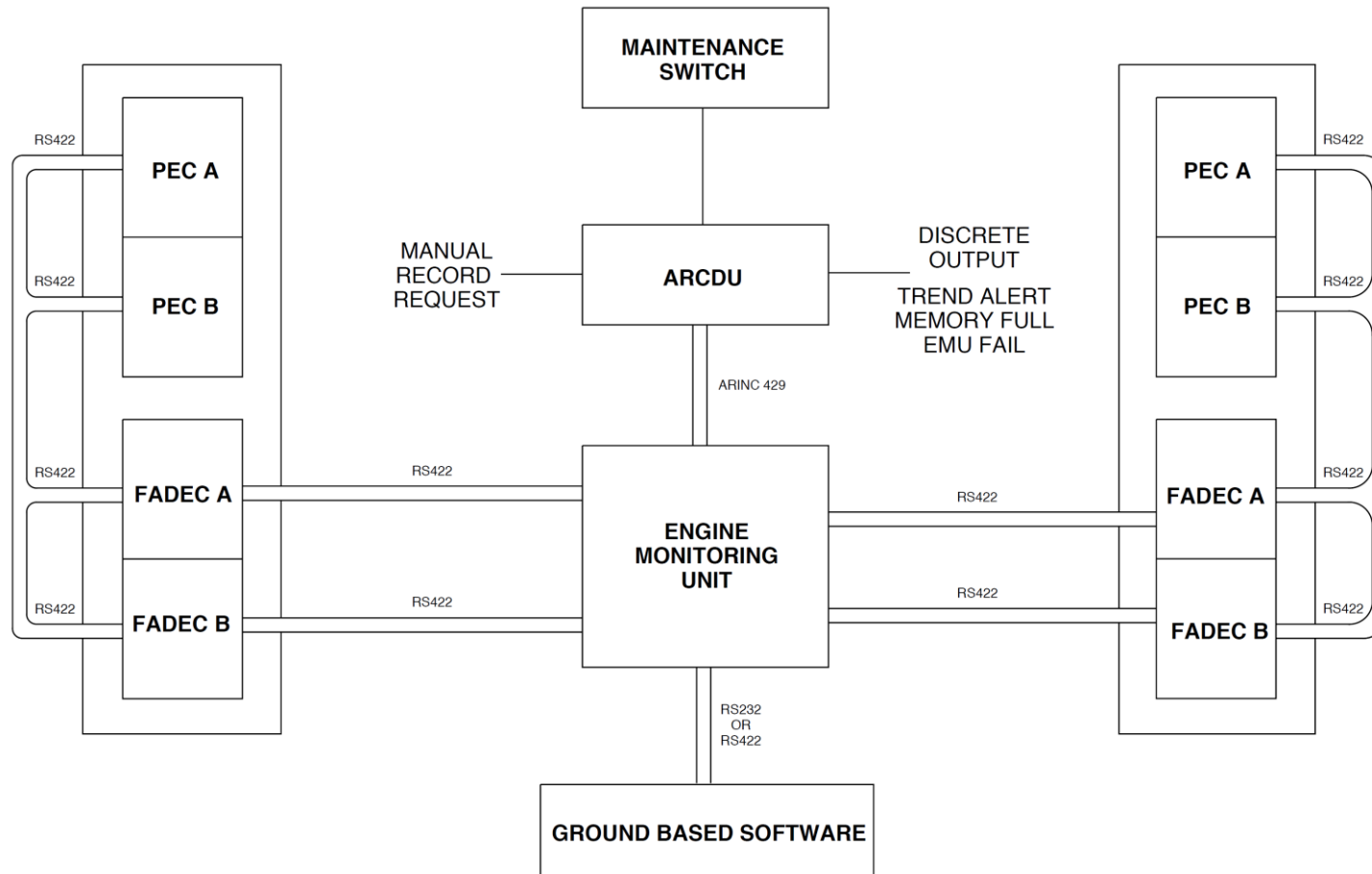


Figure - EMS Block Diagram

Detailed Description

Refer Figure - FADEC ARINC 429 Indicating System

The function of the analyzers is performed by the Engine Monitoring System (EMS) and the FADEC ARINC 429 System Each FADEC display channel (channel not in control) sends signals on an ARINC 429 bus direct to the Engine Display (ED). The FADEC information is also sent to the Integrated Flight Cabinet (IFC) for processing and retransmission to the Digital Flight Data Recorder (DFDR).

The following parameters and discretes are sent to the Flight Compartment

independent of the FADEC ARINC system:

- FADEC Caution, Fail and PEC Caution as discretes are sent directly to the Caution and Warning Panel.
- Propeller Ground Range as a discrete to the Advisory Display Unit (ADU) and then to the lights on the glareshield panel.
- Main Oil Pressure, Fuel Flow and Fuel Temperature from the engine-mounted sensors as analog signals to the IFC, and from there to the ED on an ARINC 429 DATA bus.

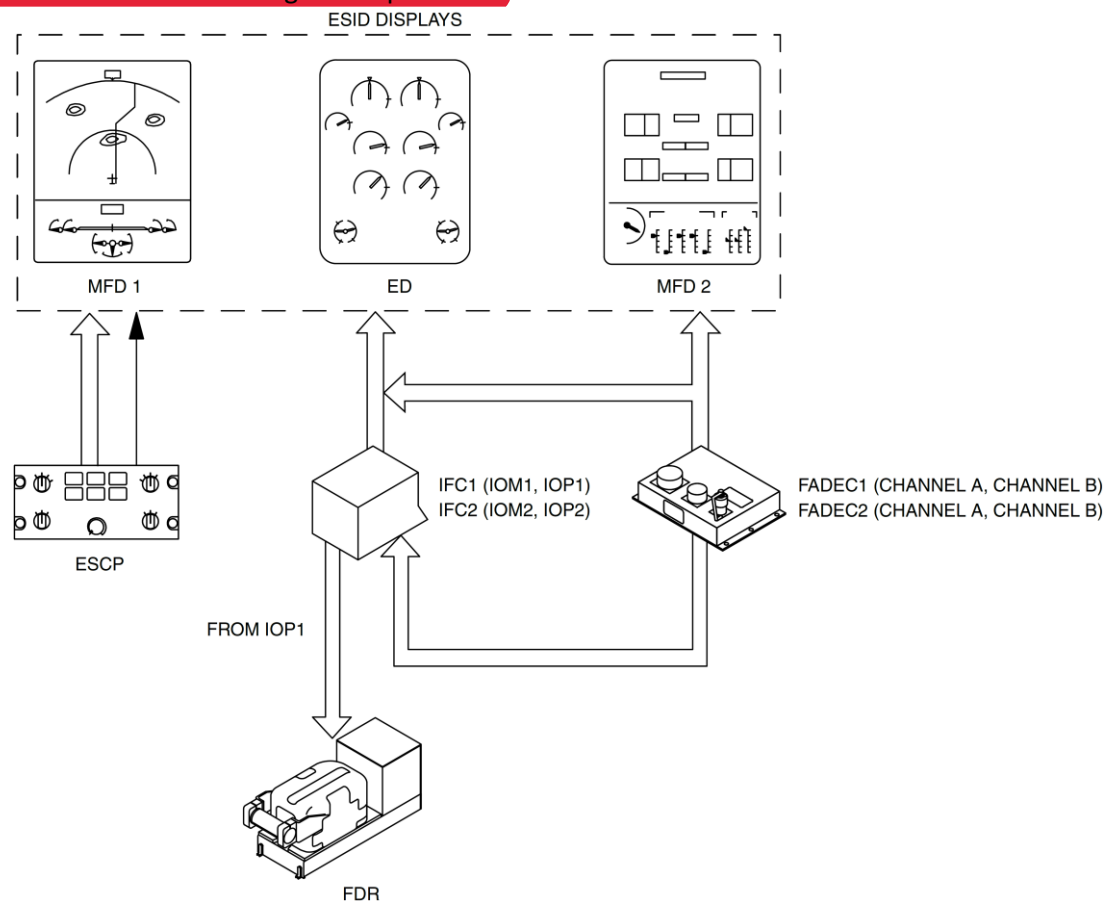


Figure - FADEC ARINC 429 Indicating System

Engine Monitoring System

Introduction

The Engine Monitoring System collects data on every significant event that occurs during the operation of the engine.

General Description

The data collected by the EMS can be accessed in the aircraft through the Audio and Radio Control and Display Unit (ARCDU) screen. The data can also be downloaded to the laptop PC based Ground Based System (GBS).

The function of the Engine Monitoring System is performed by the:

- Engine Monitoring Unit (EMU)
- Ground Based System (GBS)

The EMU is located in the flight compartment, on the side wall behind and below the First Officer's seat. The ARCDU is located in the Flight Compartment, one each side of the console behind the control quadrant.

Detailed Description

[Refer Figure - Engine Monitoring System schematic](#)

The Engine Monitoring System uses:

- The FADEC
- The Propeller Electronic Control (PEC)
- The Centralized Diagnostic System (CDS)
- The ARCDU

The EMS has the following functions:

- Snapshot and/or Trace recording in response to an engine significant event (fault code, exceedance, etc.). Where a **snapshot is a recording taken at the instant of the event and consists of approximately 79 parameters generated by the FADEC/PEC. Where a Trace is a recording initiated at the instant of the event and consists of approximately 49 parameters generated by the FADEC/PEC going back two minutes prior to the event and one minute after the event.** Resolution is one set of

parameters every 80 msec near either side of the event and 800 msec everywhere else. **A snapshot and trace can be initiated by the Flight Crew by pressing a button labelled "EVENT MARKER" on the ENGINE CONTROL PANEL.**

- Logging of Fault Codes generated by the FADEC and PEC.
- Recording of Engine Condition Trend Monitoring (ECTM) conditions automatically. Averaging of ECTM conditions for the previous Flight Hours and alert to Maintenance if the trend deviates from the norm. Recording of time to spooldown for high and low pressure spools. Alert to Maintenance if the spooldown time reduces below a minimum value significantly.
- Monitoring of Engine parameters and logging any exceedance beyond operating limitations (Ref. AMM 05-10-00). These parameters include: **ITT, Torque, NH/NL/NP, Oil Pressure, and Oil Temperature.**
- Monitoring of Engine Health discretes, and logging any change in state. These discretes include: Chip Detectors, Low Oil Pressure, Oil Filter Impending Bypass, Low Fuel Pressure, and Fuel Filter Impending Bypass.
- Step by step Power Assurance procedure. Detailed results are available following successful completion.
- Live feedback of Flight Deck engine switch state. The position of any switch which provides an input to the FADEC or PEC will be displayed. This enables the circuit to be tested quickly and easily.
- Instructions to check the operation of Engine Health Discretes.
- Confirmation of powerplant trims: Torque Gain and Bias, ITT, configuration, BETA Feedback and Power Lever Angle (PLA) Feedback.

- Review of data stored in EMU memory. This feature allows the operator to view the date/time of each recording in memory, type of recording, and Flight Deck parameters at the time of the recording.
- Summary of aircraft and FADEC configuration. Aircraft Registration, Owner and Operator can be uploaded manually. Aircraft S/N is automatically uploaded from the CMS. FADEC, PEC, and EMU S/N are logged automatically. Hours and cycles accumulated on the above is available. EMU memory usage is available.

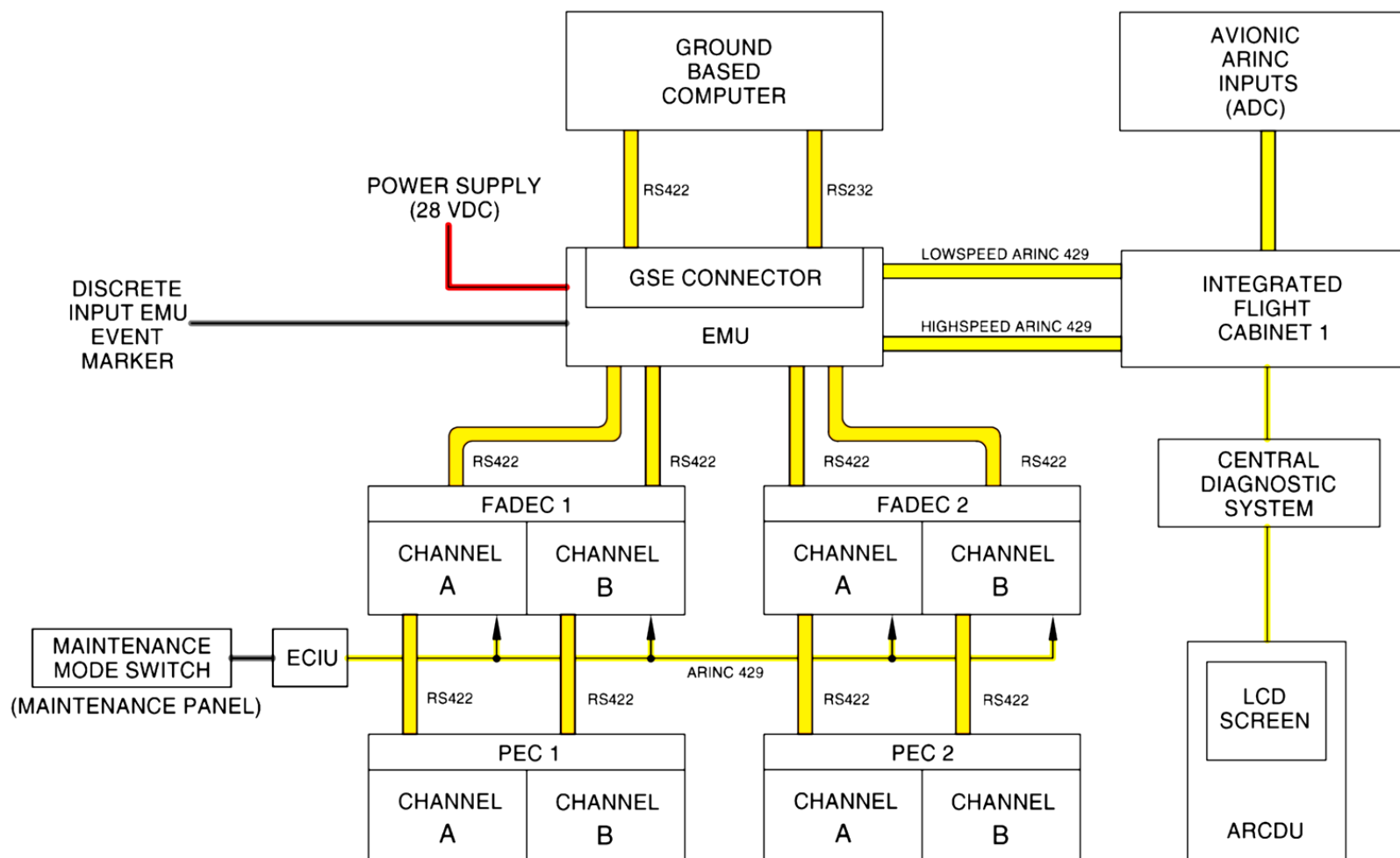


Figure - Engine Monitoring System schematic

Flight Compartment Display

Refer Figure - Engine Monitoring System, ARCDU

EMU functions can be displayed in the flight compartment using the ARCDU.

The normal mode for the ARCDU is to allow the crew to manage radio communications functions. However, in Maintenance Mode, the ARCDU becomes a display screen for the CDS.

To access the Maintenance Mode on the ARCDU, the following conditions must be satisfied:

- The aircraft is static (airspeed <50 knots (93 km/h))
- Weight is on wheels
- ARCDU GND MAINT on Central Maintenance Panel in wardrobe is selected.
- MAINT key on either or both ARCDU screens selected.

When Maintenance Mode is selected, the CMS Main Menu is displayed. The ARCDU screen has eight soft keys and three navigation keys which can be used to select menu items. To select the Engine Monitoring Unit, press the key adjacent to “EMU” on the screen.

When EMU is selected, the main menu is displayed and any EMU function can be accessed. Note that where there is an active message waiting, the menu item colour is amber rather than white. This convention is carried through the sub-menus, the sub-sub-menus, and so on.

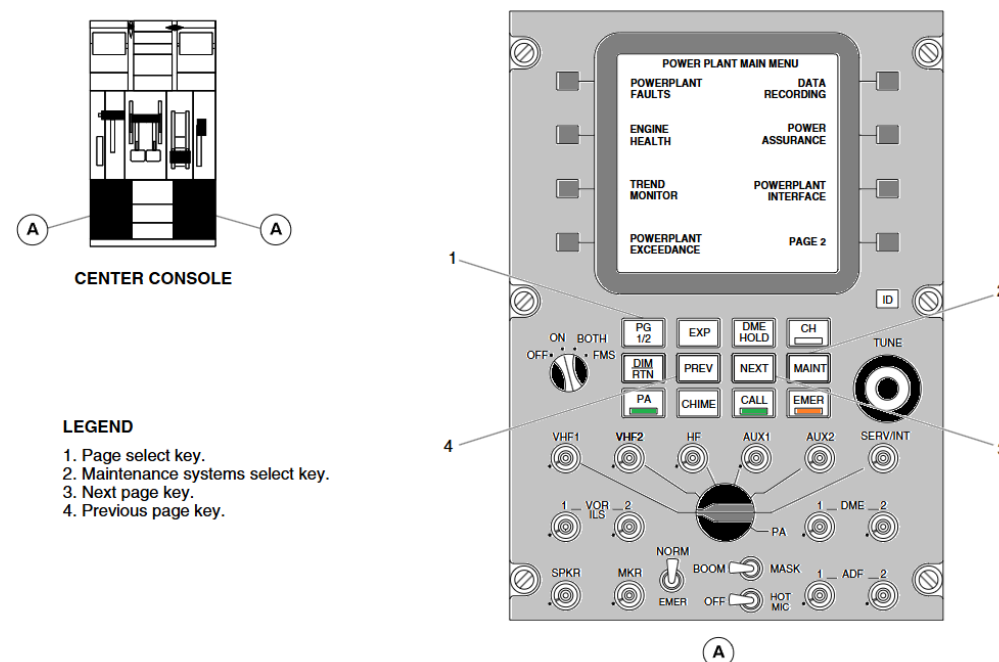


Figure - Engine Monitoring System, ARCDU

Engine Monitoring Unit (EMU)

Refer Figure - Engine Monitoring Unit

The EMU is located in the flight compartment, on the side wall behind and below the First Officer's seat.

Refer Figure - Engine Monitoring System, ARCDU

The ARCDU is located in the flight compartment, one each side of the console behind the control quadrant.

The normal mode for the ARCDU is to allow the crew to manage radio communications functions.

However, in Maintenance Mode, the ARCDU becomes a display screen for the CDS.

The data collected by the EMU can be viewed as follows:

- On aircraft on the ED Nh and torque gauges
- On aircraft on the ARCDU screen.
- The data can be downloaded to a PC GBS.

Primary fault code display is on the torque gauge and Nh gauge. Channel A, FADEC and Lane A PEC fault codes are displayed on engine torque gauge. Channel B, FADEC and Lane B PEC fault codes are displayed on Nh Gauge.

To access the Maintenance Mode on the ARCDU, the following conditions must be satisfied:

- The aircraft is static (airspeed <50 knots (93 km/h))
- Weight is on wheels
- ARCDU GND MAINT on Central Maintenance Panel in wardrobe is selected
- MAINT key on either ARCDU screens selected.

When Maintenance Mode is selected, the CMS Main Menu is displayed. The

Fault Retrieval

ARCDU screen has eight soft keys and three navigation keys which can be used to select menu items. To select the EMU, press the key adjacent to "EMU" on the screen.

When EMU is selected, the main menu is displayed and any EMU function can be accessed. Note that where there is an active message waiting, the menu item color is amber rather than white. This convention is carried through the sub-menus, the sub-submenus, and so on.

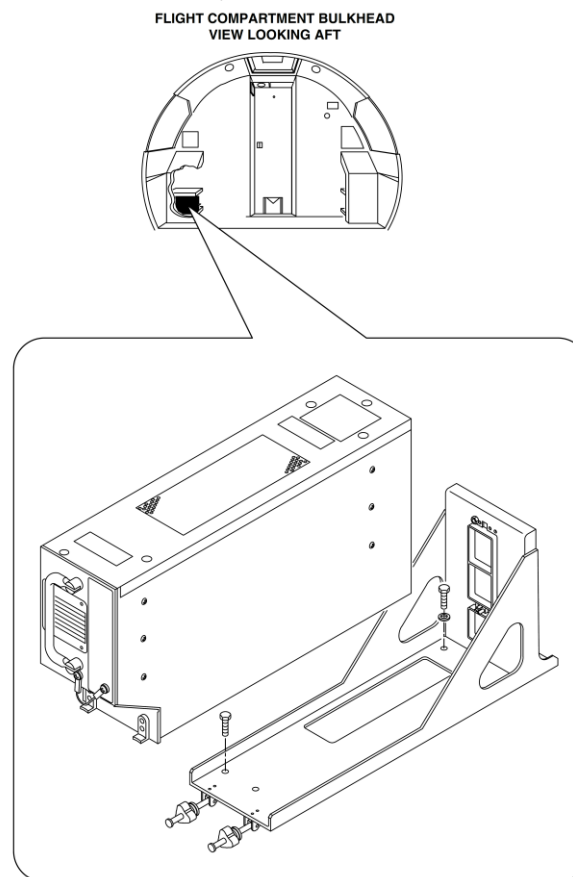


Figure - Engine Monitoring Unit